Lab Report: Experiment 2

Docker Installation, Configuration, and Running Images

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1. Objective

The objective of this lab is to:

- Pull Docker images
- Run containers
- Manage container lifecycle

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2. Procedure and Results

Step 1: Pull Image

The first step involves fetching the Nginx image from Docker Hub

- **Command:** docker pull nginx **Execution Screenshot:**

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 2: Run Container with Port Mapping

The container is started in detached mode with host port 8080 mapped to container port 80.

- **Command:** docker run -dp 8080:80 nginx

Execution Screenshot:

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker run -d -p 8080:80 nginx
96878d22f7f3869b2548292edd1356c4ecc0bd9cbff26ffef196667db0320b13
```

Step 3: Verify Running Containers

To confirm the container is active and check the assigned status and ports, the ps command is used[cite: 15].

- **Command:** docker ps

Execution Screenshot:

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker ps

CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS

96878d22f7f3   nginx     "/docker-entrypoint.…"   9 seconds ago   Up 8 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
  hardcore_hypatia
```

Step 4: Stop and Remove Container

To manage the lifecycle, the container is stopped first and then removed using its ID.

- **Stop Command:** docker stop 5ee0a6fcf8b6
- **Remove Command:** docker rm 5ee0a6fcf8b6

Stop/Remove Screenshot:

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker stop 96878d22f7f3
96878d22f7f3

saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker rm 96878d22f7f3
96878d22f7f3
```

Step 5: Remove Image

The final step in the lifecycle management is removing the image from the local repository.

- **Command:** docker rmi nginx

Execution Screenshot:

```
saharsh_kumar@LUCT573:/mnt/c/Users/Saharsh Kumar/vm-lab$ docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Deleted: sha256:5cdeff4ac3335f684128701c145cf12992f5e3669cea8ab7309257d263eb7a856bb1
Deleted: sha256:13553ab839fc3f04eb110571316ce86c00dc9a7dbd6e0960e577c7d0e94edb37
Deleted: sha256:3438c08a37f122b1e2a7a0024fefe5d904c718a3dfae622d1bded1a6cd05b6f5
Deleted: sha256:265d115060dd452fd6bd76b13990beeb7cbd4545bc82ebd13070f8b5b8b99b3c
Deleted: sha256:de7ade74c1354e0337452e80fb7198b5aa8b15ccb70d74b4272506a00e03bda02
Deleted: sha256:a40fa7f04e226ed78d693059f0223c621bc9202a95f4926a9e18ed400ca57242
Deleted: sha256:03e9a4dc9545f6a615ed07637b5e51764b6349089cb74e72e37a60d8aef4009b
Deleted: sha256:a8ff6f8cbdfd6741c10dd183560df7212db666db046768b0f05bbc3904515f03
```

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3. Result and Conclusion

Result

Docker images were successfully pulled, containers executed, and lifecycle commands performed.

Overall Conclusion

This lab demonstrated containerization using Docker, highlighting clear performance and resource efficiency. Containers are better suited for rapid deployment and microservices, whereas VMs provide stronger isolation.